DO NOW — RECALL

Operating-system types

Name: _____ Class: _____ Date: _____

1. Which scheduler uses a fixed quantum?

2. What does starvation mean?

3. What matters most to a real-time task?

4. Rising-edge flip-flop Q=0. D pulses high between edges but is 0 at next rising edge. What is new Q?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Round Robin.

2. Model answer

A task can be indefinitely denied execution.

3. Model answer

Meeting the specified response deadline.

4. Model answer

Q remains 0; only D at the active clock edge is sampled.

Vocabulary — know these terms

real-time — Deadline-bound response.

embedded — Dedicated device software.

distributed — Coordinated networked machines.

multi-user — Account-based sharing.

multitasking — Several processes progressing.

NEW LEARNING

Read, annotate and explain

Understand

Real-time OS correctness depends on deadlines, not just average response speed.

Trace

Embedded and distributed refer to system architecture and purpose; each can overlap with real-time.

Evaluate

Multi-user concerns people/account isolation, whereas multitasking concerns processes.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

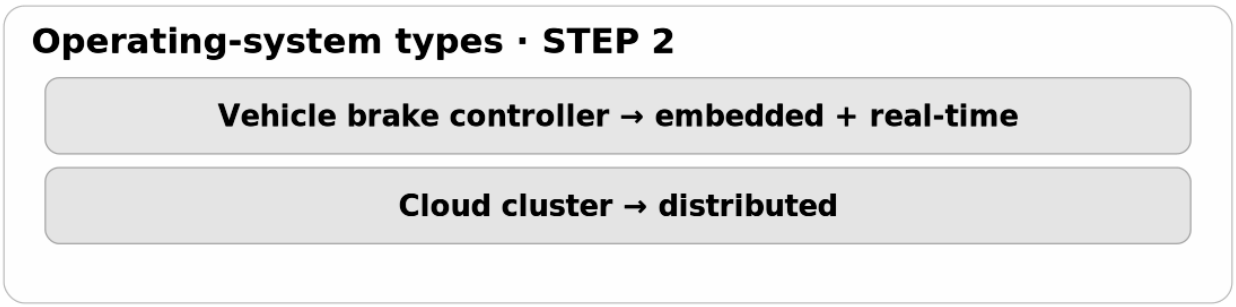
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



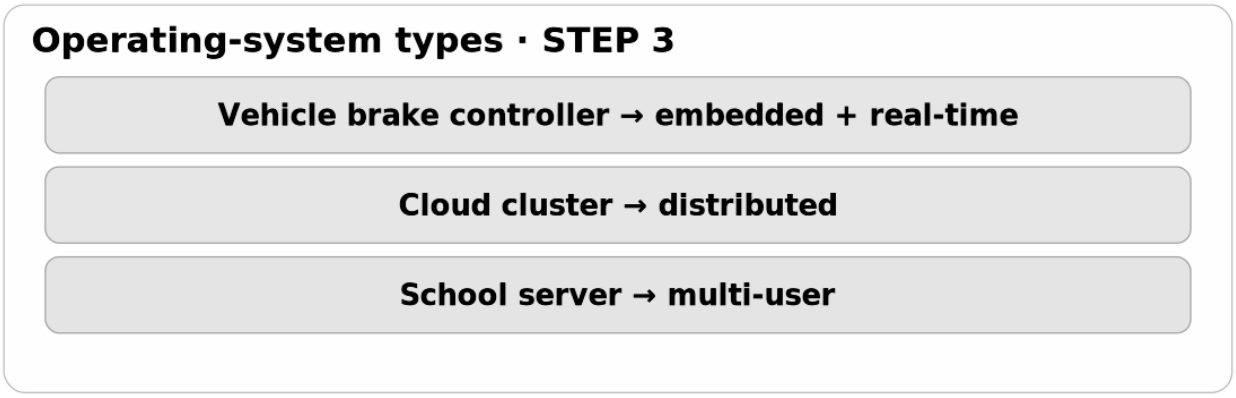
Operating-system types · STEP 1

New Learning



Operating-system types · STEP 2

New Learning

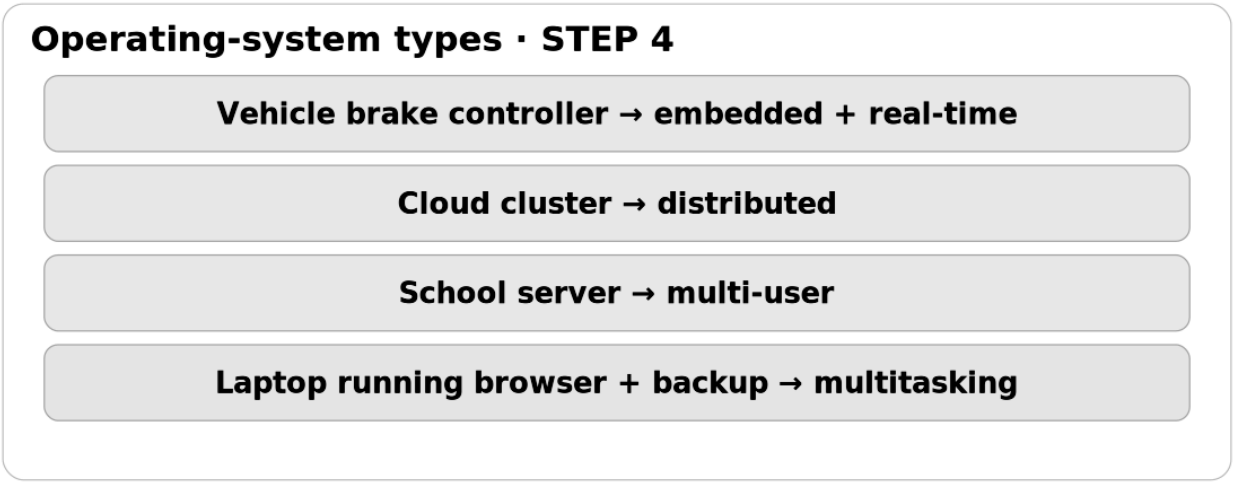


Operating-system types · STEP 3

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Operating-system types · STEP 4

I DO — WORKED EXAMPLE

Follow the worked method

A safety-critical industrial controller has a fixed response deadline. Which OS type suits it?

Predict first

Predictable bounded response is required to meet safety deadlines.

Model and check

Explain why each value follows from the earlier step.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A safety-critical industrial controller has a fixed response deadline. Which OS type suits it?

Predictable bounded response is required to meet safety deadlines.



IDO · STEP 1

A safety-critical industrial controller has a fixed response deadline. Which OS type suits it?

Predictable bounded response is required to meet safety deadlines.



IDO · STEP 2

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A university server supports many people with separate home directories. What OS characteristics?

Many accounts require permissions and several running processes share CPU time.



WEDO · STEP 1

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Many accounts require permissions and several running processes share CPU time.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Match the five OS categories to an identifying property. [5]

Question 2

Explain OS requirements of a dedicated emergency-stop controller. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why a networked cloud service can be distributed, multi-user and multitasking. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An embedded OS can never also be real-time.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A university server supports many people with separate home directories. What OS characteristics?

Many accounts require permissions and several running processes share CPU time.



WEDO · STEP 2

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and check

Explain why each value follows from the earlier step.

IMPROVE: Explain one correction to your own reasoning.